भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

Some problems on interpolation

Numerical Analysis

Solve the given problems:

1. Construct the Lagrange interpolation polynomial $p_1(x)$ of degree 1, for a continuous function $f(x)$ defined on the interval $[-1,1]$, using the interpolation points $x_0 = -1, x_1 = 1$. Show further that if the second derivative of $f(x)$ exists and is continuous on $[0,1]$, then $|f(x) - p_1(x)| \leq \frac{M_2}{2}(1 - x^2) \leq \frac{M_2}{2}, x \in [-1,1]$, where $M_2 = \max_{x\in[-1,1]} |f''(x)|$. Give an example of a function $f(x)$, and a point x, for which equality is achieved.

2. Given the distinct points $x_i, i = 0,1, \dots, n+1$, and the points $y_i, i = 0, \dots, n+1$. Let q be the Lagrange polynomial of degree n for the set points $\{(x_i, y_i): i = 0, \dots, n\}$ and let r be the Lagrange polynomial of degree n for the points $\{(x_i, y_i): i = 0, \dots, n+1\}$. Define $p(x) = \frac{(x-x_0)\, r(x)-(x-x_{n+1})q(x)}{x_{n+1}-x_0}$. Show that $p(x)$ is the Lagrange polynomial of degree $n+1$ for the points $\{(x_i, y_i): i = 0, \dots, n+1\}$.

3. In the following problems, the values of a function $f(x)$ are given. Find the interpolating polynomial that fits the data. Find an approximation to $f(x)$ at the indicated points using this polynomial.

(i)

x	-2	-1	0	1	3	4
$f(x)$	9	16	17	18	44	81

Interpolate at $x = 0.5$ and $x = 3.1$.

(ii)

x	1	3	4	5	7	10
$f(x)$	3	31	69	131	351	1011

Interpolate at $x = 3.5$ and $x = 8.0$.

4. If $f(x) = \frac{1}{x^2}$, find the divided difference $f[x_1, x_2, x_3, x_4]$.

5. Suppose that $f(x) = e^x \cos x$ is to be approximated on $[0,1]$ by an interpolating polynomial on $n+1$ equally spaced points $0 = x_0 < x_1 < x_2 < \cdots < x_n = 1$. Determine n so that the truncation error will be less than 0.0001 in this interval.

6. If $f(x) = e^{ax}$, show that $\Delta^n f(x) = (e^{ah} - 1)^n e^{ax}$.

7. Construct the interpolating polynomial that fits the data

x	0	0.1	0.2	0.3	0.4	0.5
$f(x)$	-1.5	-1.27	-0.98	-0.63	-0.22	0.25

using the Newton forward or backward difference interpolation. Hence estimate the values of $f(x)$ at $x = 0.15, 0.25,$ and 0.45.

8. Using the Newton's backward difference interpolation, construct the interpolating polynomial that fits the data

x	0.1	0.3	0.5	0.7	0.9	1.1
$f(x)$	-1.699	-1.073	-0.375	0.443	1.429	2.631

Estimate the value of $f(x)$ at $x = 0.6$ and $x = 1.0$.

9. The following data represents the function $f(x) = e^x$.

x	1	1.5	2.0	2.5
$f(x)$	2.7183	4.4817	7.3891	12.1825

Estimate the value of $f(2.25)$ using the Newton's forward difference interpolation and Newton's backward difference interpolation. Compare with the exact value. Obtain the bound on the truncation error.

10. The following data represent the function $f(x) = \cos(x+1)$.

x	0.0	0.2	0.4	0.6
$f(x)$	0.5403	0.3624	0.1700	-0.0292

Estimate $f(0.5)$ using the Newton's backward difference interpolation. Compare with the exact value. Obtain the bound on the truncation error.

11. Obtain the piecewise quadratic interpolating polynomials for the function $f(x)$ defined by the data

x	-3	-2	-1	1	3	6	7
$f(x)$	369	222	171	165	207	990	1779

Hence find an approximate value of $f(-2.5)$ and $f(6.5)$.

12. Construct the Hermite interpolation polynomial that fits the data

x	$f(x)$	$f'(x)$
0	0	1.0
0.5	0.4794	0.8776
1.0	0.8415	0.5403

Estimate the value of $f(0.75)$. Find a bound on the error. If the data represents the function $f(x) = \sin x$, then find the actual error at $x = 0.75$.

13. Construct the Hermite interpolation polynomial that fits the data

x	$f(x)$	$f'(x)$
2	29	50
3	105	105

Interpolate $f(x)$ at $x = 2.5$. Now fit the cubic polynomial $P(x) = \sum_{i=0}^{3} c_i \, x^i$, for $c_i \in \mathbb{R}$. to the given data in this problem. Are these polynomials same?

14. In the following problems, obtain the piecewise quadratic interpolating polynomials for the function $f(x)$ defined by the given data. Interpolate at $x = -0.5$ and 2.0

x	-2	0	1	3	4
$f(x)$	-23	1	4	82	193

15. Obtain the piecewise cubic interpolating polynomials for the function $f(x)$ defined by the data

x	-5	-4	-2	0	1	3	4
$f(x)$	275	94	-334	-350	-349	-269	-94

Hence find an approximate value of $f(-3.0)$ and $f(2.0)$.

16. Obtain a quadratic spline fit, $s(x)$, to the function $f(x) = x^4$on the interval $-1 \leq x \leq 1$, corresponding to the partition $x_0 = -1, x_1 = 0, x_2 = 1$ and satisfying the condition $s'(-1) = f'(-1)$.

17. Obtain the cubic spline fit for the data

x	0	1	2	3
$f(x)$	1	4	10	8

under the end conditions $f''(0) = 0 = f''(3)$ and valid in the interval [1,2]. Hence estimate of $f(1.5)$. Draw a rough sketch of the solution curves.

18. Find the values of α and β such that the function

$$f(x) = \begin{cases} x^2 - \alpha\, x + 1, & 1 \leq x \leq 2, \\ 3\, x - \beta, & 2 \leq x \leq 3 \end{cases}$$

is a quadratic spline.

19. (a) Show there is a unique polynomial $p(x)$ for which

$p(x_0) = f(x_0),\ p(x_2) = f(x_2),\ p'(x_1) = f'(x_1),\ p''(x_1) = f''(x_1)$, where $f(x)$ is a given function and $x_0 \neq x_2$. Derive a formula for $p(x)$.

(b) Let $x_0 = -1,\ x_1 = 0,\ x_2 = 1$. Assuming $f(x)$ is four times continuously Differentiable on [−1,,1], show that for $-1 \leq x \leq 1$,

$f(x) - p(x) = \frac{x^4-1}{4!} f^4(\xi_x)$, for some $\xi_x \in [-1,1]$.

20. Consider the problem of finding a quadratic polynomial $p(x)$ for which $p(x_0) = y_0, p'(x_1) = y_1',\ p(x_2) = y_2$ with $x_0 \neq x_2$ and $\{y_0, y_1', y_2\}$ the given data. Assuming that the nodes x_0, x_1, x_2 are real, what conditions must be satisfied for such a $p(x)$ to exist and be unique ?